

PIPING ISOMETRICS MANAGEMENT SYSTEM (PIMS) USER MANUAL

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Document Control

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1. INTRODUCTION

The Piping Isometrics Management System (PIMS) is a comprehensive web-based application designed to facilitate and streamline the complete lifecycle of isometric drawing management within engineering projects. PIMS provides a centralized platform for all stakeholders—including modellers, process checkers, material checkers, stress analysts, GL (Group Lead) reviewers, and SGL (Sub-Group Lead) coordinators—to collaborate efficiently on piping isometric preparation, checking, and issuance.

The system has been developed to integrate data from multiple sources including 3D modeling software, project management databases, and material control systems into a single unified user interface. This integration allows project engineers and coordinators to track the progress of isometric drawings, manage document revisions, coordinate reviews, incorporate feedback from multiple checkers, and plan timely release of final isometrics to the field.

PIMS is built with a focus on simplicity, modularity, and user-friendliness. The system is designed to minimize data entry requirements through intelligent data import capabilities while providing comprehensive reporting and tracking features that give users complete visibility into project status at any time.

2. FEATURES

The PIMS package includes the following key features:

- **Web-Based Architecture:** PIMS is a modern web application accessible through standard internet browsers, requiring no client-side software installation
- **Multi-User Role Support:** The system supports multiple user roles with role-based access control including Modeller, Process Checker, Material Checker, Stress Checker, GL, SGL, and HOD (Head of Department)
- **Modular Design:** PIMS is organized into self-contained modules that can be accessed independently based on user requirements
- **Intuitive User Interface:** Modules are designed to be self-explanatory with clear navigation and minimal training requirements
- **Relational Database Management:** PIMS utilizes a robust RDBMS architecture to maintain data integrity, consistency, and security while avoiding data redundancy
- **Large Volume Data Handling:** The system is optimized to handle large volumes of project data including thousands of isometric drawings and associated metadata
- **Single Point Data Entry:** PIMS implements a one-time data entry approach where information entered once is automatically propagated across relevant modules
- **Comprehensive Reporting:** The system provides extensive reporting capabilities with export functionality to standard formats (Excel, PDF) for further analysis and documentation
- **Real-Time Progress Tracking:** Users can monitor progress of isometric drawings through all stages of the workflow in real time
- **Document Version Control:** PIMS maintains complete revision history for all drawings with tracking of changes and approvals

3. SYSTEM REQUIREMENTS

HARDWARE REQUIREMENTS

- Minimum Desktop Computer with Intel Core i5 or higher processor (or equivalent AMD)
- Windows 10 or higher Operating System (or equivalent Linux/macOS)
- Minimum 4 GB of Random Access Memory (RAM)
- Minimum 50 GB of Hard Disk Capacity for data and program files

- Standard laser printer (for report printing)
- Internet connectivity (for web-based access to PIMS)

SOFTWARE REQUIREMENTS

- **Web Browsers:** Google Chrome 90+, Mozilla Firefox 88+, Microsoft Edge 90+, or Safari 14+
- **PDF Viewer:** Adobe Acrobat Reader DC or equivalent
- **Office Applications:** Microsoft Excel 2016+ or LibreOffice Calc (for report editing)
- **Network:** Active Internet or Intranet connectivity at minimum 10 Mbps
- **VPN:** If accessing from remote locations, appropriate VPN client for secure connection

4. BRIEF SYSTEM DETAILS

PIMS is engineered to serve the comprehensive requirements of piping engineers and project coordinators in managing isometric drawings through the complete project lifecycle. The system integrates information from various sources and compiles them into a single, cohesive user interface that provides clear and actionable information.

4.1 System Architecture

PIMS is structured as a layered web application consisting of:

- **Presentation Layer:** User-friendly web interface built with responsive design
- **Application Layer:** Business logic and workflow processing engine
- **Data Layer:** Secure database management with backup and recovery capabilities

4.2 Key Modules

The PIMS system includes the following primary modules:

A. Authentication & Dashboard Module

This is the initial entry point where users provide their credentials and gain access to the system. The dashboard provides an overview of assigned tasks and key metrics.

B. User Management Module

Enables HOD to create jobs, assign roles to team members, and manage project access. SGL users can add or remove team members from assigned projects and manage unit-wise role assignments.

C. Isometric Upload Module

Modellers can upload PDF isometric drawings with automatic parsing of filenames according to PIMS convention. The module supports drag-and-drop interface for easy file handling and maintains version control.

D. Review & Checking Module

Supports multi-stage checking workflow where Process Checkers, Material Checkers, and Stress Checkers can review drawings, provide comments, and recommend revisions. The system tracks reviewer assignments and completion status.

E. Workflow Management Module

Manages the complete workflow from initial upload through final approval. Tracks the status of each drawing as it moves through the checking stages, maintains audit trail of all changes, and provides notifications for pending actions.

F. Reporting Module

Provides comprehensive dashboards and reports showing:

- Isometric workflow status and progress
- AFC (Authority For Construction) issuance status
- Stress analysis critical lines
- Checker and reviewer performance metrics
- Project-wide progress summaries

G. PDF Viewer Module

Integrated PDF viewer for online document review without requiring external applications.

5. OPERATING DETAILS

5.1 Accessing PIMS

To access PIMS, users must navigate to the web application URL provided by the system administrator. Once the application loads, the user is presented with the login screen.

Login Screen

The login page displays the PIMS system logo and prompts the user for authentication credentials. Users must enter:

- **Employee ID:** The unique identifier assigned by the organization
- **Password:** The secure password assigned during account creation

The system displays a "Sign-In" button to submit credentials. If incorrect credentials are entered, an error message appears prompting the user to try again.

Upon successful authentication, the user is redirected to the main dashboard.

5.2 Dashboard & Navigation

After login, users are presented with the PIMS dashboard which serves as the central hub for all activities. The dashboard layout includes:

- **Header Section:** Displays PIMS logo, version number, current user information, and logout option
- **Left Navigation Panel:** Provides access to all functional modules based on user role
- **Main Content Area:** Displays the selected module or default welcome message
- **Top Menu Bar:** Contains quick access buttons for common functions

5.3 Role-Based Access

PIMS implements strict role-based access control. The user's assigned roles determine which modules and functions they can access:

Modeller Role

- Upload isometric drawings
- View assigned drawings
- Access PDF viewer
- Submit drawings for checking

Process Checker Role

- View assigned drawings for process checking
- Provide process comments
- Accept or reject drawings
- Track checking status

Material Checker Role

- Review drawings for material specifications
- Verify material lists and specifications
- Provide material comments
- Approve or request revisions

Stress Checker Role

- Review drawings for stress analysis compliance
- Verify support specifications
- Provide stress analysis feedback
- Approve critical line drawings

GL (Group Lead) Role

- Review completed checking work
- Provide group-level approval
- Access comprehensive reports
- Monitor team performance

SGL (Sub-Group Lead) Role

- Manage team member assignments
- Assign drawings to checkers
- Monitor checking progress
- Escalate issues and manage timelines

HOD (Head of Department) Role

- Create new projects
- Assign SGLs to projects
- Manage overall project setup
- Access system-wide reports and analytics

5.4 Main Menu Navigation

The left panel contains collapsible menu sections:

Inbox

- **My Tasks:** Displays drawings assigned to the current user for action
- **Final Isometrics:** Shows completed isometric drawings ready for issuance
- **Rejected Isometrics:** Lists drawings rejected during checking process
- **Notifications:** Displays system notifications and alerts

Project Documents

- **My Projects:** Projects where user has active assignments
- **Other Projects:** Other projects visible to the user
- Displays hierarchical tree of projects with expandable units and zones

5.5 My Tasks Module

The My Tasks section displays drawings awaiting action by the current user. The interface shows:

Column	Description
Job No	Project identifier
Unit No	Unit within project
Line No	Specific line number
Revision	Current revision number of drawing
Stress Critical	Indicates if drawing requires stress analysis
From	Originating role/user
Date Sent	When drawing was sent for review
Claimed Roles	Roles to which drawing is assigned

Users can filter tasks by job number, unit, line number, or status. Available actions include "Claim" (accept assignment), "View", and "Comment".

5.6 Isometric Upload Module

The Upload Module allows modellers to submit isometric drawings to PIMS.

File Naming Convention

PIMS uses a specific filename format for all uploads to ensure proper parsing and tracking:

```
SERVICE-UNIT-ISONUM-ZONE.pdf
```

Example: ABD-OIC-7501-D.pdf

Where:

- **SERVICE:** Service code (e.g., ABD, AI, AP, CBD, FL)
- **UNIT:** Unit code (e.g., OIC, OIS, OID)
- **ISONUM:** Isometric number (e.g., 7501, 7502, 6601)
- **ZONE:** Zone designation (single letter, e.g., A, B, C, D, E)

Upload Process

1. Click "Upload Isometrics" button from the main dashboard
2. Select project from dropdown menu
3. Click in the file drop area or browse to select PDF files
4. Multiple files can be selected at once
5. System automatically parses filenames and validates format
6. Click "UPLOAD FILES" button to submit
7. System displays upload progress and confirmation

Upload Status Tracking

The system provides real-time feedback during upload:

- Files successfully uploaded display "uploaded successfully" status
- Failed uploads show error messages explaining the issue
- A download log button becomes available after upload completion
- Users can download the upload log as an Excel file for their records

5.7 My Tasks Detail View

When a user selects a task from the My Tasks list, the system displays detailed information including:

- **Task Header:** Shows Job Number, Unit, Zone, Line Number, Revision, Criticality status
- **Claimed Roles:** Displays roles already assigned to this task
- **Claimed Role Details:** Shows who claimed each role and when
- **PDF Viewer:** Displays the isometric drawing for review
- **Comments Section:** Shows previous comments from other reviewers
- **Current Role Assignment:** Allows user to claim or perform their assigned role

5.8 Drawing Review and Checking

When reviewing a drawing, users can:

1. **View the Drawing:** Use the integrated PDF viewer to examine all details
2. **Review Previous Comments:** Read feedback from earlier checkers
3. **Add Comments:** Enter specific comments or issues found
4. **Select Comment Type:** Choose from predefined categories (Process, Material, Stress, etc.)
5. **Indicate Action:** Select "Approve", "Revision Required", or "On Hold"
6. **Submit Review:** Post comments and move drawing to next stage

5.9 Final Isometrics Module

The Final Isometrics section displays drawings that have completed all checking stages and are ready for:

- Final approval and signature
- Release to field teams
- Archive and documentation

Users can:

- View final drawing details
- Download approved isometric copies
- Access complete revision history
- Generate issuance certificates

5.10 Reports Module

PIMS provides comprehensive reporting capabilities through multiple dashboard views:

Isometric Workflow Status

Shows count of drawings in each workflow stage:

- Uploaded in PIMS
- Checking Under Progress
- AFC Issued
- Unclaimed (waiting for checker assignment)

AFC ISO Issuance Status

Displays distribution of drawings by revision:

- Rev 0 (first issue)
- Rev 1 (first revision)
- Rev 2 (second revision)
- Rev 3 and higher

Stress Analysis Critical Lines

Shows status of stress-critical drawings:

- Total critical lines
- Completed analysis
- In process analysis

Checker/Reviewer Performance

Tracks performance metrics for each team member including:

- Lines processed
- Average processing time
- Quality metrics
- Approval/rejection rates

Detailed Line Reports

Filterable reports showing individual drawing information:

- Line number and unit
- Zone information
- Current status
- Assigned checkers and their status
- Dates of key milestones

Export Options

All reports can be exported in:

- Excel format (.xlsx) for further analysis
- PDF format for distribution and archiving

5.11 User Management (HOD Console)

The HOD (Head of Department) console provides administrative functions:

Create Job

1. Navigate to User Management → For HOD → Create Job
2. Enter project details
3. Select applicable units for project
4. Define project parameters and constraints
5. Save project configuration

Assign SGL

1. Select project from list
2. Choose units to include
3. Select employees to assign as SGL
4. Confirm assignments

5. Selected employees become SGL for specified units

5.12 Role Management (SGL Console)

The SGL (Sub-Group Lead) console enables:

Add Members

1. Select project and units
2. Browse list of available employees
3. Assign roles to selected employees:
 - Modeller
 - Process Checker
 - Material Checker
 - Stress Checker
 - GL (Group Lead)
 - SGL (Sub-Group Lead)
4. Confirm assignments

Remove Members

1. Select project and units
2. View current assignments
3. Select employees to remove
4. Confirm removal of specified roles
5. Member loses access to assigned drawings

5.13 Stress Critical Index Upload

SGLs can upload a stress critical index file to identify which lines require stress analysis:

1. Navigate to Misc. → Upload Stress Critical Index
2. Select the stress index file (Excel format)
3. System parses file and identifies critical lines
4. Critical lines marked in all subsequent reports and workflows

5.14 Project Documents Navigation

Users can browse project structure through the hierarchical tree view:

- Projects are displayed at top level
- Expanding a project shows its units
- Units can be expanded to show zones
- Zones show individual drawings/isometrics
- Right-clicking on items shows context menu with options

6. ERROR MESSAGES

The following error messages may be displayed by PIMS:

Authentication Errors

Error Message	Cause	Resolution
Invalid Employee ID or Password	Incorrect credentials entered	Verify employee ID and password. Contact system administrator if forgotten
Authentication Required	Session expired or not authenticated	Log in again with valid credentials
Access Denied	User does not have permission	Contact system administrator to verify role assignments

File Upload Errors

Error Message	Cause	Resolution
Invalid Filename Format	PDF filename does not match PIMS convention	Rename file to SERVICE-UNIT-ISONUM-ZONE.pdf format
File Not Selected	No file chosen for upload	Click Browse button and select a PDF file
Non-PDF File Selected	Uploaded file is not in PDF format	Ensure file is in PDF format before uploading
Another User is Currently Uploading	Concurrent upload conflict	Wait a few minutes and try again
File Size Exceeds Limit	PDF file is too large	Reduce file size or split into multiple files

Project & Unit Errors

Error Message	Cause	Resolution
Please Select a Job Number	No project selected	Select project from dropdown menu
Please Select at Least One Unit	No units selected	Select one or more units
Units Not Available for Selected Project	Project has no configured units	Contact SGL or HOD to configure units
Failed to Load Projects	System cannot retrieve project list	Refresh page or contact system administrator

Role & Permission Errors

Error Message	Cause	Resolution
Modeller Not Authorised for the Unit	User's role not permitted for drawing location	Contact SGL to verify role assignments
Insufficient Permissions	User role cannot perform action	Contact system administrator
Cannot Claim Role - Already Assigned	Another user already claimed this role	Wait for that user to complete or contact SGL

Data Processing Errors

Error Message	Cause	Resolution
Error While Fetching User Data	System cannot retrieve user information	Refresh page, log out and log back in
Error While Fetching Units Data	System cannot retrieve unit list	Check network connection and try again
Error While Updating Selection	System cannot save selections	Try again with fewer selections
Error While Binding Data to Grid	Display data processing failure	Refresh page or reduce filtered data range
Error While Generating Excel File	Report export failure	Try exporting smaller data sets first

Comment & Feedback Errors

Error Message	Cause	Resolution
Cannot Post Comment - Drawing Locked	Drawing is currently being edited by another user	Wait for other user to finish or contact SGL
Comment Cannot Be Empty	User attempted to submit blank comment	Enter descriptive comment before posting
File Not Attached to Comment	Required file attachment missing	Attach required document to comment
Cannot Upload File - Unsupported Format	Attached file type not supported	Use supported formats (PDF, PNG, JPG, Excel)

System & Server Errors

Error Message	Cause	Resolution
Connection Timeout	Server did not respond within expected time	Check network connection and try again
Session Expired	User session has timed out due to inactivity	Log in again to continue
Internal Server Error	Unexpected system error occurred	Note error details and contact system administrator
Database Connection Error	Cannot connect to database	Wait a moment and try again. Contact administrator if persists
Service Temporarily Unavailable	System maintenance or overload	Wait and try again later

Report Errors

Error Message	Cause	Resolution
No Data Available for Selected Criteria	Filters returned no matching records	Modify filter parameters to broaden search
Error Generating Report	Report processing failed	Simplify report parameters and try again

Error Message	Cause	Resolution
Error Exporting to Excel	Excel file generation failed	Try exporting smaller data sets or simpler reports
Filter Not Applied	Selected filters not working	Clear filters and try again

QUICK REFERENCE GUIDE

Common Tasks

To Upload an Isometric Drawing

1. Click "Upload Isometrics"
2. Select Project
3. Select PDF file (follow naming format)
4. Click Upload Files

To Review a Drawing

1. Click "My Tasks"
2. Select drawing to review
3. View PDF using embedded viewer
4. Add comments
5. Select action (Approve/Revision Required)
6. Submit

To View Project Reports

1. Click "Reports"
2. Select reporting parameters
3. Select Job Number and Unit(s)
4. Click "Fetch Report"
5. View on-screen or export to Excel

To Manage Team Members (SGL only)

1. Click "User Management" → "For SGL"
2. Select "Add Members"
3. Choose project and units
4. Assign roles to employees
5. Click "Assign Roles"

Keyboard Shortcuts

Shortcut	Action
Ctrl + S	Save current changes
Ctrl + P	Print current view/report
Ctrl + E	Export to Excel

Shortcut	Action
F5	Refresh page
Esc	Close dialog or menu

Support & Contact

For technical support or to report issues:

- **System Administrator:** Contact ITS help desk
- **Training & Documentation:** Refer to this manual
- **Bug Reports:** Use the bug report form in application
- **Feature Requests:** Contact project management team

Document Revision History

Version	Date	Changes
1.0	Nov 2025	Initial release

End of Document

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